

TABLE III
Internal vs. External Generalization and Calibration

Metric	Internal	External	Change
Accuracy	99.95%	37.97%	-61.97 pp
Macro F1	99.94%	30.10%	-69.84 pp
Balanced Accuracy	99.92%	34.90%	-65.02 pp
Mean Confidence	99.97%	92.83%	-7.14 pp
Confidence-Accuracy Gap	0.0234 pp	54.8602 pp	+54.84 pp
Expected Calibration Error	0.000558	0.548602	+0.548044
Brier Score	0.001101	1.140233	+1.139132
Negative Log-Likelihood	0.005038	5.455558	+5.450520

Note. Internal results correspond to the strict shared-eight PlantVillage comparison subset (n = 1,938); external results correspond to the mapped PlantDoc test set (n = 79). Performance and calibration values are from the locked PART 20 analysis. Change values preserve the locked full-precision results; displayed Internal and External endpoints are rounded. ECE = Expected Calibration Error; NLL = Negative Log-Likelihood; pp = percentage points. No post-hoc calibration, external fine-tuning, threshold optimization, or model reselection was performed.